

Open Source

Lesson 45 — Source Code and Licensing Models

JKSSB Computer Awareness

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What Is Source Code?

- ▶ **Source code** is the set of instructions written by a programmer in a programming language such as C, Java, or Python.
- ▶ It is **human-readable**: a person can open it, read it, and change it.
- ▶ A **compiler** or **interpreter** converts source code into the machine-readable form the computer runs.
- ▶ The machine-readable form is called **object code** or the executable; ordinary users rarely see it.

The one idea

Whoever holds the source code controls how the software can be studied, changed, and shared.

Open Source vs Proprietary

Point	Open source	Proprietary
Source code	Available to the public	Kept secret (closed source)
View / modify	Allowed, subject to the licence	Not allowed without permission
Redistribute	Allowed, subject to the licence	Not allowed
Typical user gets	Source code + executable	Only the compiled executable

The deciding factor is the source code, not the price.

What “Open Source” Allows

- ▶ **View** the source code: anyone may read how the software works.
- ▶ **Modify** it: a user may change it, fix bugs, or add features.
- ▶ **Share** it: copies and modified versions may be redistributed.
- ▶ These permissions are granted by the software's **licence**, so the exact terms differ from one project to another.

Exam key (PYQ-26)

An **open-source licence** lets users view, change, and share the source code. That three-part meaning is the tested idea.

Proprietary (Closed-Source) Software

- ▶ **Proprietary software** keeps its source code **secret**.
- ▶ The user receives only the **compiled executable**, not the source.
- ▶ Copying, modifying, or redistributing it is not allowed without the owner's permission.
- ▶ Examples: **Microsoft Windows**, **Apple macOS**, **Microsoft Office**, **Adobe Photoshop**, **Oracle Database**.

Freeware Is Not Free Software

- ▶ **Freeware** is software given away **free of cost**, but its source code is usually **not** provided.
- ▶ **Free software** is about **freedom**, not price: the user is free to study, change, and share it.
- ▶ “Free” in free software means “free as in freedom”, not “free as in free beer”.
- ▶ A program can be freeware and still be proprietary.

Trap alert (TRAP-23)

“Free” in **freeware** means **zero cost**; “free” in **free software** means **liberty**. Do not treat the two as the same.

Free Software and Its Four Freedoms

- ▶ The **Free Software Foundation (FSF)** was founded by **Richard Stallman** to promote software freedom.
- ▶ Freedom 0: run the program for any purpose.
- ▶ Freedom 1: study how it works and change it.
- ▶ Freedom 2: redistribute copies to help others.
- ▶ Freedom 3: distribute modified versions to the public.

Keep it straight

Free software, open source, and freeware are three different ideas. Only the first two are about source-code freedom.

Public Domain Software

- ▶ **Public domain** software has **no copyright** owner.
- ▶ Anyone may use, modify, and redistribute it **without any restriction**.
- ▶ No licence is needed, because no exclusive rights remain.
- ▶ This differs from open source: open-source software is still copyrighted, but a licence grants the freedoms.

Definition

Public domain means the work belongs to everyone; open source means the work is owned but licensed for public use.

Commercial and Shareware Software

- ▶ **Commercial software** is developed and **sold for profit**. It is often proprietary, but not always.
- ▶ **Shareware** is distributed as a **free trial**: it works for a limited period or with limited features.
- ▶ To keep using shareware after the trial, the user must **pay**.
- ▶ Shareware is not open source: the source code stays hidden.

Trap alert (TRAP-23)

Open source, freeware, and shareware are **not** the same. Open source shares the **source**; freeware is free of **cost**; shareware is a **trial**.

Repositories: Where Code Lives

- ▶ A **repository** (repo) stores a project's source code and every change made to it.
- ▶ **Version control** records the history, so older versions can be restored.
- ▶ **Git** is the most widely used distributed version-control system.
- ▶ Popular hosting sites: **GitHub**, **GitLab**, **Bitbucket**, and **SourceForge**.

Community Development

- ▶ Open-source projects are built by a **community** of contributors around the world.
- ▶ A **fork** is a contributor's own copy of the project for experimenting or improving.
- ▶ A **pull request** proposes changes back to the main project for review.
- ▶ **Maintainers** review changes, and an **issue tracker** records bugs and feature requests.

Why it matters

Many hands can inspect and improve the code, which is why community review is treated as a strength of open source.

Basic Licensing Concepts

- ▶ **Copyright** gives the creator exclusive rights by default.
- ▶ A **licence** is the legal permission that says what others may do with the software.
- ▶ **Copyleft** licences require modified versions to stay under the **same** licence.
- ▶ **Permissive** licences allow reuse in other projects, including proprietary ones, with few conditions.

Remember

The licence, not the price, decides whether software is truly open source.

Copyleft vs Permissive Licences

Type	Examples	Main condition
Copyleft	GNU GPL	Derivative works must use the same licence
Permissive	MIT, Apache 2.0, BSD	Reuse allowed with licence notice; few limits

GPL = GNU General Public License. Apache 2.0 adds an explicit patent grant.

Common Open-Source Licences

- ▶ **GNU GPL** — GNU General Public License; a strong copyleft licence.
- ▶ **MIT License** — very permissive; requires the licence notice to be kept.
- ▶ **Apache License 2.0** — permissive, with an explicit patent grant.
- ▶ **BSD License** — permissive, with minimal conditions.
- ▶ **Creative Commons** — used for creative works, not usually for software.

Well-Known Open-Source Software

- ▶ Operating systems and kernels: **Linux**, **Android**.
- ▶ Web browsers: **Mozilla Firefox**; **Chromium** (the open-source base of Google Chrome).
- ▶ Office and productivity: **LibreOffice**.
- ▶ Media and graphics: **VLC media player**, **GIMP**, **Blender**.
- ▶ Servers and languages: **Apache HTTP Server**, **MySQL**, **Python**, **Git**, **WordPress**.

Many are free of cost as well, but it is the open source code, not the price, that defines them.

PYQ Focus: Android and the Licence Test

- ▶ **PYQ-24**: the open-source **mobile operating system** is **Android**.
- ▶ Android is built on the open-source **Linux** kernel.
- ▶ **PYQ-26**: an open-source licence allows users to **view, change, and share** the source code.
- ▶ By contrast, Microsoft Windows and Apple macOS are proprietary, and iOS is a proprietary mobile OS.

Takeaway

If the question names a mobile OS, **Android** is the open-source answer.

Trap Alert: Three Confused Terms

Term	What is free	Source code?
Open source	Use, study, change, share	Available
Freeware	Cost only	Usually hidden
Shareware	A limited trial	Hidden

Trap alert (TRAP-23)

Do not write “open-source = freeware = shareware”. They differ in what is free and whether the source code is available.

Lesson 45: Recall

- ▶ **Source code** is human-readable; a compiler or interpreter turns it into machine code.
- ▶ **Open source** lets users view, change, and share the source code, subject to its **licence**; **proprietary** keeps the source secret.
- ▶ **Freeware** is free of **cost**; **free software** is about **freedom** (FSF, Richard Stallman, four freedoms).
- ▶ **Public domain** has no copyright; **shareware** is a paid trial; **commercial** software is sold for profit.
- ▶ **Repositories** (Git, GitHub, GitLab) and **community development** (forks, pull requests, maintainers) drive open source.
- ▶ **GPL** is copyleft; **MIT**, **Apache 2.0**, and **BSD** are permissive.
- ▶ **Android** is the open-source mobile OS (PYQ-24); open source is not freeware or shareware (TRAP-23).